



Automated Classification of Different Categories of Assamese Songs Using KNN and MLP

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Abstract. This work is a tribute to the musical heritage of Assam that showcases an automated system for the classification of the regional Assamese songs of the state by the use of machine learning techniques beyond the traditional ones. By using KNN and MLT, which uses advanced audio feature extraction methods like Mel-Frequency Cepstral Coefficients (MFCCs) and spectrograms, to classify Assamese songs into music genres such as Bihu, Kamrupiya Lokageet, Goalporiya Lokageet, Borgeet, and Naam. This initiative is an endeavor to take over the situation where challenges faced due to modernization and globalization are the causes of the destruction of such melodious music.

Besides that, the digital archive of the cultural wealth of Assam acts as a bridge to connect with the residents of the world to raise awareness of intercultural appreciation and research. Technological advancements have indeed become a mighty force in the preservation and promotion of cultural and musical traditions. To name one, this repository is a necessary source of healing for scholars and educators, empathy and cultural sensitivity for diagnosis

Keywords: Assamese songs, K-Nearest Neighbors(KNN), Multi-Layer Perceptron(MLP), Mel-Frequency Cepstral Coefficients (MFCCs), spectrograms, Bihu, Borgeet, Kamrupiya Lokageet, Goalporiya Lokageet, Naam.

1. Introduction

Assam, a state in northeastern India, is known for its vibrant and diverse musical tradition of the state.[1] The local songs of the state were the products of

the people living there, and the songs represented the cultural, social, and historical narratives of the communities. The songs were the main part of the Assamese identity.[2] The songs also include various genres like Bihu, Kamrupiya Lokageet, Goalporiya Lokageet, Borgeet, Naam, and many more, each with its own dance tempo, melody, and lyric content.[3] It is of utmost importance that the songs are kept alive and also that those songs are made available to the people from different parts of the world to keep the cultural fabric of Assam intact.[4]

Deep learning techniques are used by our project to fulfill this aim of preservation and accessibility as well as being the means.[5] Deep learning, which is a subfield of artificial intelligence, is most suitable for the identification of patterns and carrying out classifications based on complicated data.[6] The utilization of deep learning computer programs, which are the algorithms of artificial intelligence, on voice recordings of Assamese songs, enables us to make a system that correctly recognizes the input songs and classifies them according to their locational and genre features.[7]

After the features have been extracted, they are passed to machine learning models, particularly to a K-Nearest Neighbor (KNN) and Multi-Layer Perceptron (MLP)[12]. KNN and MLP fit this work perfectly since they are capable of uncovering intricate patterns in the data, and therefore, they are the best for audio processing[13]. KNN and MLP are trained on the data set, hence, it learns to associate some features with some genres of Assamese songs[14]. Through constant training and tuning, the model improves its accuracy and finally arrives at a high degree of certainty in the classification of the songs[15].

Beyond the technical achievement of song classification, our project has broader implications for research and education[16]. By creating a comprehensive and accessible repository of classified Assamese songs, we provide a valuable resource for scholars and researchers in ethnomusicology, cultural studies, and related fields[17]. This repository can serve as a basis for further studies on Assamese music, facilitating a deeper understanding of its evolution, diversity, and cultural significance[18].

Moreover, our project promotes cross-cultural appreciation and education[19]. By making the rich musical heritage of Assam accessible to a global audience, we encourage greater recognition and appreciation of this unique cultural tradition[20]. Educational institutions can use our classified dataset and automated system as teaching tools, introducing students to the diverse sounds and stories of Assam[21].

1.1 CATEGORIAL INTRODUCTION:

1. Bihu: Bihu is a vibrant folk festival and musical tradition in Assam, celebrated with dance, music, and rituals marking seasonal changes. The three primary Bihu festivals—Bohag, Magh, and Kati Bihu—celebrate spring,

harvest, and prosperity, respectively. Bihu songs, lively and rhythmic, capture the spirit of Assamese life.

2. Kamrupiya Lokageet: These are folk songs from the Kamrup region of Assam, reflecting the cultural and social ethos of the area. Kamrupiya Lokageet encompasses themes like love, nature, and local traditions, often performed with simple musical accompaniments to emphasize the lyrical content.

3. Goalporiya Lokageet: Originating from the Goalpara district of Assam, Goalporiya Lokageet showcases the diverse linguistic and cultural heritage of the region. These folk songs blend Assamese and Bengali elements, conveying everyday experiences, love, and spirituality in a distinctive lyrical style.

4. Borgeet: Borgeet are classical devotional songs composed by the saint-scholar Sankardev and his disciple Madhavdev in the 15th and 16th centuries. Performed within Assam's Vaishnavite monasteries, these songs express deep devotion to Lord Krishna, characterized by structured melodies and spiritual lyrics.

5. Naam: Naam is a form of congregational chanting and prayer in Assam's Vaishnavite tradition, particularly within the Namghar (community prayer hall). These chants, performed collectively, express devotion and reinforce communal bonds, fostering unity and spiritual connection among participants.

1.2 Outline: Assam, a state in the northeastern part of India, is the musical landscape of the area with its expressive and varied musical tradition, which, in turn, reflects the cultures, societies, and historical narratives of its communities[22]. The singing of the different regions in the state, Bihu, Kamrupiya Lokageet, Goalporiya Lokageet, Tokari Geet, Naam, etc., are an integral part of Assamese identity, and all these six songs are different from each other in rhythm, melody, and lyrical content[23].

Our work, which uses deep learning methods to bring more people to these songs and to preserve them, has been successful through these technologies, which are best suited to recognize the patterns in complex data. This is followed by the collection and digitization of an eclectic dataset of Assamese regional songs, feature extraction such as Mel-Frequency Cepstral Coefficients (MFCCs), and spectrograms to represent the main auditory elements of the songs.

The Convolutional Neural Network (CNN) is the one to which these features are submitted, and it undergoes training on the dataset to determine the classification of the given songs as per their genre. Hence, those classifications of songs, which this framework accomplishes with a significant level of exactness as well as the subsequent collection of songs that plays an essential role in research and education, are now made available without much difficulty.

Academics belonging to such disciplines as ethnomusicology and cultural studies may gain from the deepening of their insight into the variety and the development of Assamese music by utilizing the repository. Moreover, this project is not only helpful for teachers and students in Assam who can learn

from the repository about the music of the state but also the admirers of the region's diverse culture across the world.

1.3 Purpose: This project is all about saving the musical heritage of Assam that is diverse in nature by using deep learning techniques for classification and categorization of local songs. By creating a system that can automatically analyze and identify the common characteristics of different music genres of Assamese, this project is intended to develop a platform that is easily accessible to the scholars, teachers, and the whole world community. As a result, this not only makes the research in ethnomusicology and cultural studies possible but also facilitates the research and furthers the cross-cultural exchange.

1.4 Objectives:

The fundamental goals that have been targeted are:

- i Collect six different categories of Assamese local songs and create a comprehensive dataset.
- ii. Label the data with care to ensure accuracy.
- iii. Audio data preprocessing to be done in depth including feature extraction and normalization
- iv. Implementation of Machine Learning and Deep Learning models for classification, along with designing.

2. LITERATURE SURVEY

Teng Li [12], in the study, "Music genre classification is achieved by transforming music into computer-understandable data using Mel-Frequency Cepstral Coefficients (MFCC), capturing rhythm and timbre. Subsequently, two Convolutional Neural Networks (CNNs) are utilized, and each one is improved with Black Hole Optimization to attain the highest possible accuracy of genre identification. The results of these CNNs that are very well adjusted are amalgamated, metaphorically, as the combined scent of two bloodhounds following different paths to corner prey, resulting in a system whose accuracy in genre classification exceeds 95%. This high accuracy can be employed in music recommendation systems, and more developments in optimization algorithms or neuromorphic computing can make this method suitable for large-scale applications.

Yigang Meng [13], in the study, "this paper experimentally contrasts two different machine learning methodologies for music genre classification, namely Convolutional Neural Networks (CNNs) and XGBoost. XGBoost proves to be extremely good at unearthing patterns hidden in structured data like Mel-Frequency Cepstral Coefficients (MFCCs) that depict the rhythm and timbre of a song. On the contrary, CNNs dissect the Mel spectrograms, i.e. graphics of the sound frequencies, to differentiate genres.

The empirical work shows that XGBoost using MFCC features does better than CNNs which implicates that for audio data in the form of tables like MFCCs, XGBoost may be a more viable option. This underlines the proficiency of MFCCs in detecting the timbral and rhythmic aspects of music, which serve as the recognizable patterns for XGBoost. Subsequent research might consider novel feature extraction methods for the music industry other than MFCCs and Mel spectrograms that could reflect one's music taste in terms of harmony or emotion. Genre classification tools could become more developed with the features such as subgenres and real-time functionalities that could facilitate new approaches to music analysis and recommendation systems."

Krishnarjun Bora, Manash Pratim Barman and Arnab N. Patowary [14], in the study, the approach uses features extracted from audio recordings, like Mel-frequency cepstral coefficients (MFCCs) and spectrograms, to capture an instrument's timbre and pitch.

Bora, K. et al. [5]. Author scientist authors in their work [5], used four different supervised learning methods to classify musical melodies with the help of Mel-frequency Cepstral Coefficients (MFCC). These methods mentioned were :Decision Tree Classifier (DTC), Linear Discriminant Analysis (LDA), Random Forest Classifier, and Support Vector Machine (SVM). The outcome of the model evaluation led to the accuracies as follows.

Decision Tree Classifier (DTC) recorded 73.58%, Linear Discriminant Analysis (LDA) attained 85.58%, Random Forest Classifier scored 86.11%, and Support Vector Machine (SVM) was the best with 94.17%. These numbers were computed from a total of 250 samples (50 of each folk music type), and based on them, SVM was the most successful classifier among the four.

Jothimani, S., and K. Premalatha.[6]. Jothimani, S., and K. Premalatha.[6], The MFF-SAUG algorithm led to a breakthrough in speech-based emotion recognition over various datasets, with impressive accuracy as follows: 92.6% for RAVDESS, 89.9% for CREMA, 84.9% for SAVEE, and an outstanding 99.6% for TESS. The union of all datasets resulted in an accuracy of 94.5% of the total. These findings are a testament to the performance of the new model over the old CNN + LSTM model.

Chakrabarty, S.,The authors have described a system for extracting similar music patterns, mainly using the Correlation Coefficient. The algorithm described in the paper additionally involves creating a mobile app for recommending the other songs from Indian Folk Music (IFM) for the listeners of Indian Classical Music (ICM).

Ke. Xu[8], employed Cross-Entropy Cost Function: This function was used to quantify the divergence between the predicted values and the target values in the output layer of the neural network, where genre probabilities are estimated using the softmax function.

Adam Optimization Algorithm: The optimization tool used as the basis for this work was Adam, which is an adaptive learning rate algorithm that efficiently modifies the learning rates for different parameters by computing the first-order and second-order moment estimations of gradients.

3. PROPOSED METHOD

The proposed problem definition, dataset used and different methods used to extract features specifically required for this task are as below:

A. *Problem Definition*

In the digital age, Assamese songs, cherished treasures of Assam's cultural heritage, face pressing challenges in preservation. These orally transmitted melodies, steeped in linguistic nuances and melodic richness, encounter hurdles in systematic documentation, leaving manual classification susceptible to errors. This project addresses this critical issue by harnessing the power of deep learning to automate the precise recognition and categorization of various Assamese songs sourced from audio recordings.

A. *Dataset preparation*

The dataset consists of 2500 Assamese regional songs collected from online platforms, classified into five different categories. For rigorous model training, the dataset is thoughtfully partitioned: 80% for training, 10% for validation, and 10% for testing. All files are in the .wav format, ensuring uniformity for effective deep learning analysis and reflecting a diverse and representative collection of Assamese musical heritage.

B. *Preprocessing*

In the preprocessing function, audio files are loaded and processed for uniformity and suitability in deep learning analysis. The function, implemented with the librosa library, standardizes the sampling rate to 16,000 Hz and limits the duration to 30 seconds. To accommodate variations in audio length, padding is applied, creating a consistent matrix structure. This meticulous preprocessing ensures that each audio file is transformed into a format conducive to subsequent model training, enhancing the robustness of the classification system.

C. *Feature Extraction*

The feature extraction process in this project, facilitated by the Librosa library, incorporates several key parameters to ensure the creation of informative and standardized representations of the Assamese Regional songs. A specified sample rate of 16,000 Hz, denoted by `target_sampling_rate=16000`,

establishes a consistent measure of audio signal per second during processing. Additionally, the duration parameter, set to 30 seconds ($\text{duration}=30$), enforces uniformity in the length of audio clips for analysis and model training. Mel-frequency cepstral coefficients (MFCCs), computed with $n_mfcc=40$, play a pivotal role in capturing the unique spectral features of each audio signal. This amalgamation of parameters contributes to a comprehensive feature set, allowing the deep learning model to discern intricate patterns and nuances within the Assamese Bihu folk songs for accurate classification during subsequent training and testing phases.

D. Framework

Audio Library in use is librosa and deep frameworks in use are Keras and Theano.

E. Constituent Algorithms: The different algorithms used are:

- i. KNN: K-Nearest Neighbors (KNN) is a versatile algorithm used in both classification and regression tasks in machine learning. Its non-parametric nature means it doesn't make assumptions about data distribution and instead stores all training data for decision-making during prediction. When given a new data point, KNN calculates the distance to all existing points in the training set and selects the k -nearest neighbors, where k is a predefined constant. For classification, it assigns the most common class among these neighbors to the new point, while for regression, it calculates the average (or another statistic) of the target values. One critical consideration is choosing the appropriate value of k , as it influences the model's sensitivity to noise and potential overfitting or underfitting. While computationally intensive, KNN is straightforward to understand and implement, making it a popular choice for beginners and as a baseline model in machine learning workflows.

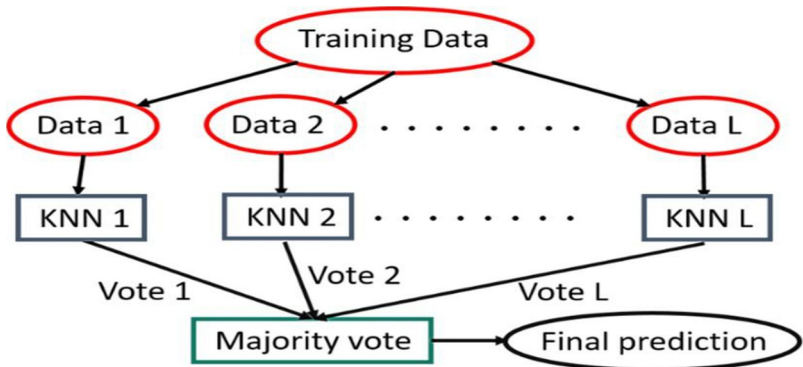


Fig 1: KNN Model Architecture

ii. MLP: A multilayer perceptron (MLP) is a type of artificial neural network characterized by its layered architecture, consisting of an input layer, one or more hidden layers, and an output layer. Each layer comprises interconnected nodes, or neurons, which process and transmit information through weighted connections. In an MLP, information flows forward from the input layer, undergoing nonlinear transformations through the hidden layers before producing an output. Through a process called back-propagation, the network adjusts the weights of its connections during training to minimize the difference between predicted and actual outputs, thereby learning to approximate complex functions and patterns within data. MLPs are widely used in machine learning for tasks such as classification, regression, and pattern recognition, owing to their flexibility, scalability, and capability to model nonlinear relationships in data.

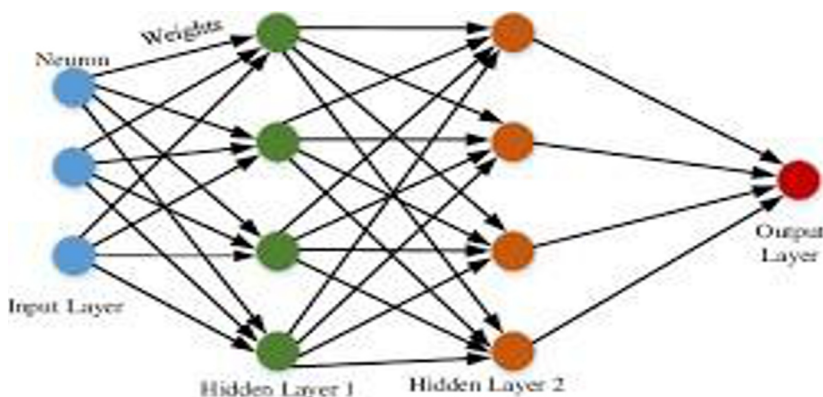


Fig 2: MLP Model Architecture

F. Model Training

The training phase involves preparing an extensive dataset comprising Assamese Bihu and non-Bihu folk songs. The audio data undergoes preprocessing, merging Bihu and non-Bihu samples and adapting them to suit the model architecture's requirements. The model is then constructed, incorporating a base structure and additional layers for effective classification.

Over a series of 10 training epochs, the model adjusts its internal parameters using the Adam optimizer and minimizing categorical

cross entropy loss. This iterative training refines the model's capacity to recognize intricate patterns within Assamese Bihu folk songs. Following training, the model's performance is assessed on multiple datasets—training, validation, and test sets—using accuracy as the key metric. This training workflow seamlessly integrates audio feature extraction, transfer learning, and model training to achieve accurate and robust classification of Assamese folk songs.

G. Prediction

In the prediction phase, the trained model applies its learned knowledge to classify new, unseen audio samples as either Assamese Bihu or non-Bihu folk songs. Leveraging the intricate patterns and features it acquired during training, the model assesses the unique characteristics of each input audio file. The prediction process involves generating a probability distribution across the defined classes, ultimately assigning a label to the input based on the highest probability.

The model's predictive accuracy is crucially evaluated on real-world data, ensuring its effectiveness in making accurate classifications beyond the training and validation sets. Through this predictive capability, the model demonstrates its practical utility in automatically identifying and categorizing Assamese folk songs, contributing to the preservation and appreciation of the rich cultural heritage embodied in Bihu music.

H. System Implementation

We delve into the practical implementation of our proposed algorithms, covering the steps involved in building and deploying machine learning (ML), deep learning (DL), and hybrid models. We provide a detailed account of our methodology, from data preprocessing to model training, and finally, the integration of the trained models into a live environment.

We will meticulously explain each model, step by step, detailing the preprocessing steps, training process, and integration into a live environment. This chapter provides a comprehensive description of the practical implementation of our proposed algorithms, highlighting the intricacies involved in building and deploying models across different paradigm.

4. Results, Observations and Comparative Analysis

We will conduct a comprehensive analysis of the results obtained from the application of both the algorithms. Our objective is to scrutinize their performance and behavior, evaluating the accuracy they achieved on the training, validation, and testing sets within the dataset. The model performance results indicate that the Multi-Layer Perceptron (MLP) obtained accuracy with 95.59% followed by K-Nearest Neighbors (KNN) with an accuracy of 94.76%.

Precision Scores and Confusion Matrices

i. KNN

Accuracy: 0.81

Classification Report:

	precision	recall	f1-score	support
0	0.86	0.89	0.87	1021
1	0.62	0.95	0.75	1000
2	0.95	0.83	0.89	985
3	0.86	0.77	0.81	989
4	0.92	0.61	0.74	1005
accuracy			0.81	5000
macro avg	0.84	0.81	0.81	5000
weighted avg	0.84	0.81	0.81	5000

I. fig: Confusion Matrix for KNN

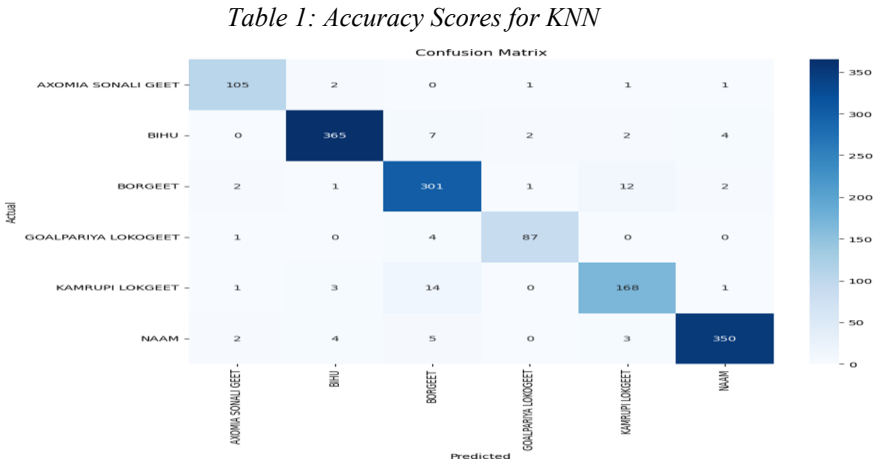


Fig 3: Confusion Matrix for KNN

	Precision	Recall	F1-Score
Bihu	86.00%	89.00%	87.00%
Borgeet	62.00%	95.00%	75.00%
Goalpariya Lokageet	95.00%	83.00%	89.00%
Kamrupiya Lokageet	86.00%	77.00%	81.00%
Naam	92.00%	61.00%	74.00%

Table 2: Detailed Scores Table for KNN

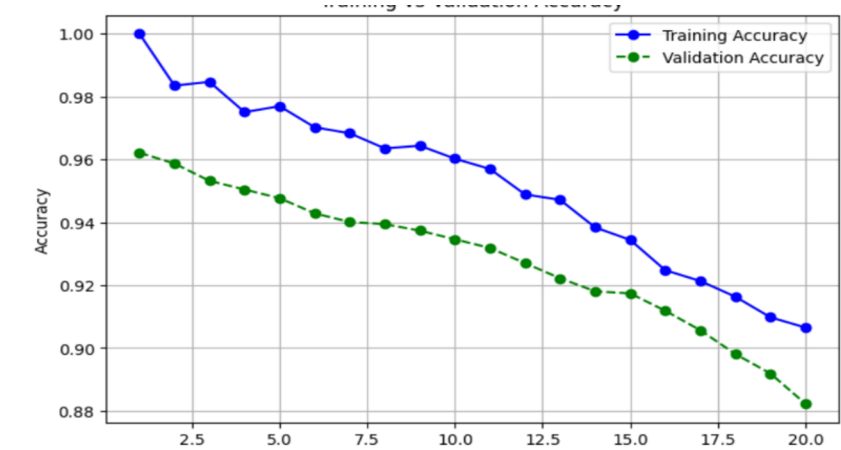


Fig 4: Graphical Representation of Accuracy Scores

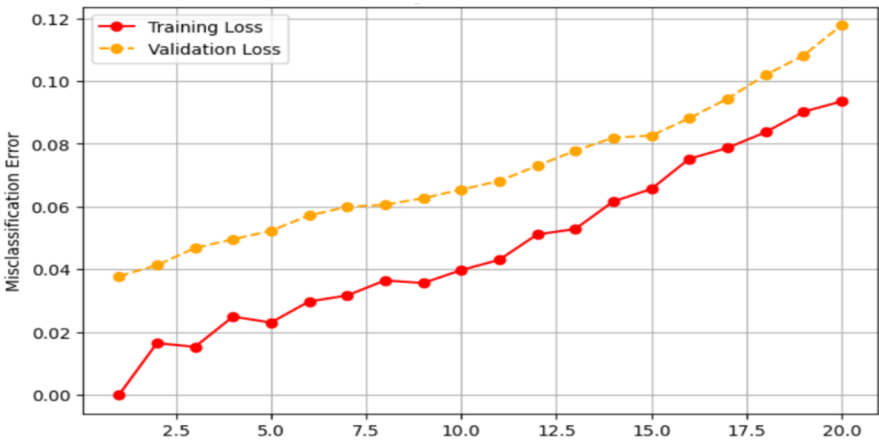


Fig 5: Graphical Representation of Loss Scores

i. MLP

Accuracy: 0.8038

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.87	0.86	1021
1	0.78	0.78	0.78	1000
2	0.85	0.90	0.87	985
3	0.77	0.74	0.75	989
4	0.76	0.73	0.74	1005
accuracy			0.80	5000
macro avg	0.80	0.80	0.80	5000
weighted avg	0.80	0.80	0.80	5000

Table 3: Accuracy Scores for MLP

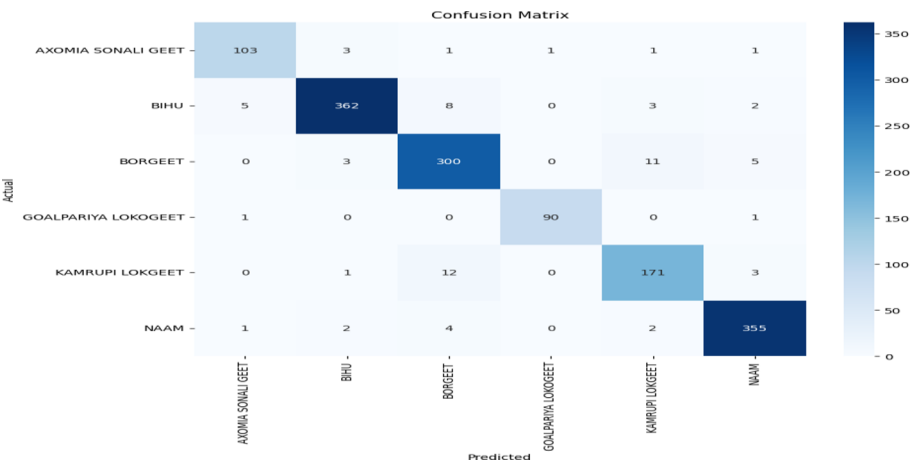


Fig 6: Confusion Matrix for MLP

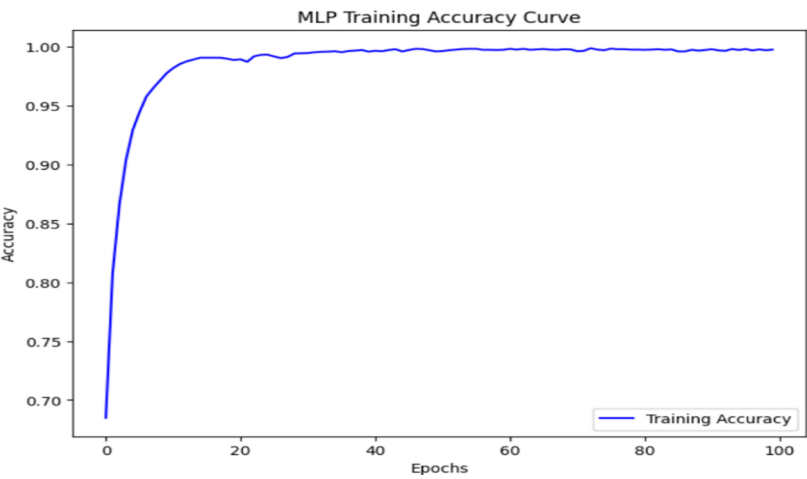


Fig 7: Graphical Representation of Accuracy Scores

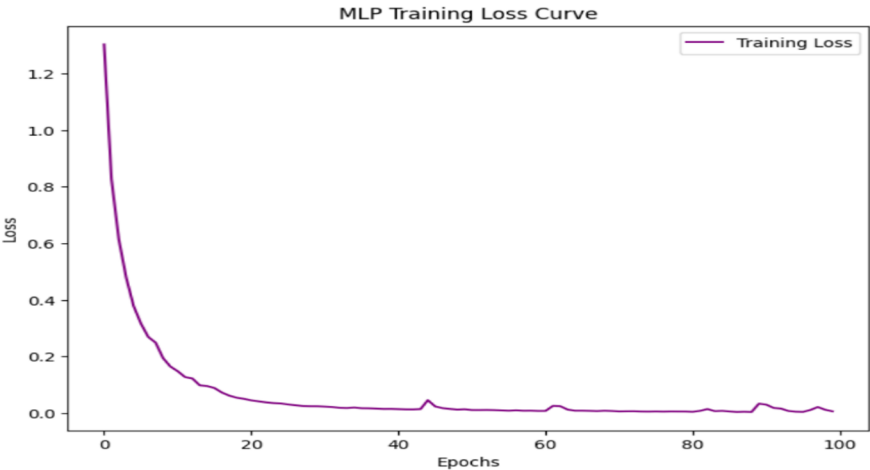


Fig 8: Graphical Representation of Loss Scores

	Precision	Recall	F1-Score
Bihu	85.00%	87.00%	86.00%
Borgeet	78.00%	78.00%	78.00%
Goalpariya Lokageet	85.00%	90.00%	87.00%
Kamrupiya Lokageet	77.00%	74.00%	75.00%
Naam	76.00%	73.00%	74.00%

Table 4: Detailed Scores Table for MLP

5. Conclusion

In conclusion, To sum up, this work has effectively employed machine learning techniques to maintain and propagate the musical heritage of Assam by creating an automated classification system for regional songs. Using K-Nearest Neighbor (KNN) and Multi Layer Perceptron (MLP) along with advanced feature extraction methods, we have gotten together the digital archive which very well indicates the different genres of Assamese music. This new method confronts the problems of saving music caused by change and globalization, thus it is ensured that these traditional songs will be there for the coming generations. Besides, this project is a major resource for the research and the education sector as it can spread the appreciation of the cultural diversity of Assam across the world. By the help of technology and cultural preservation, we are willing to ignite the fire of pride.

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